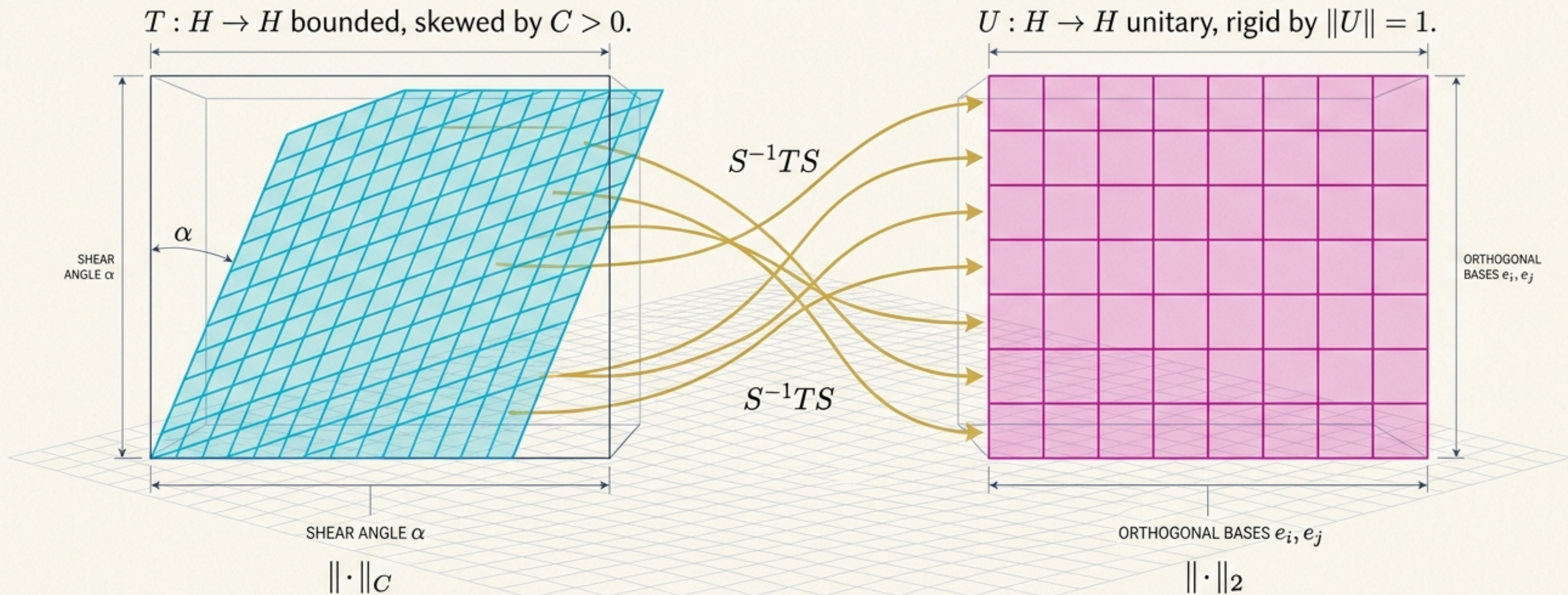


Straightening the Skew

The 2026 Resolution of Kadison's Similarity Conjecture

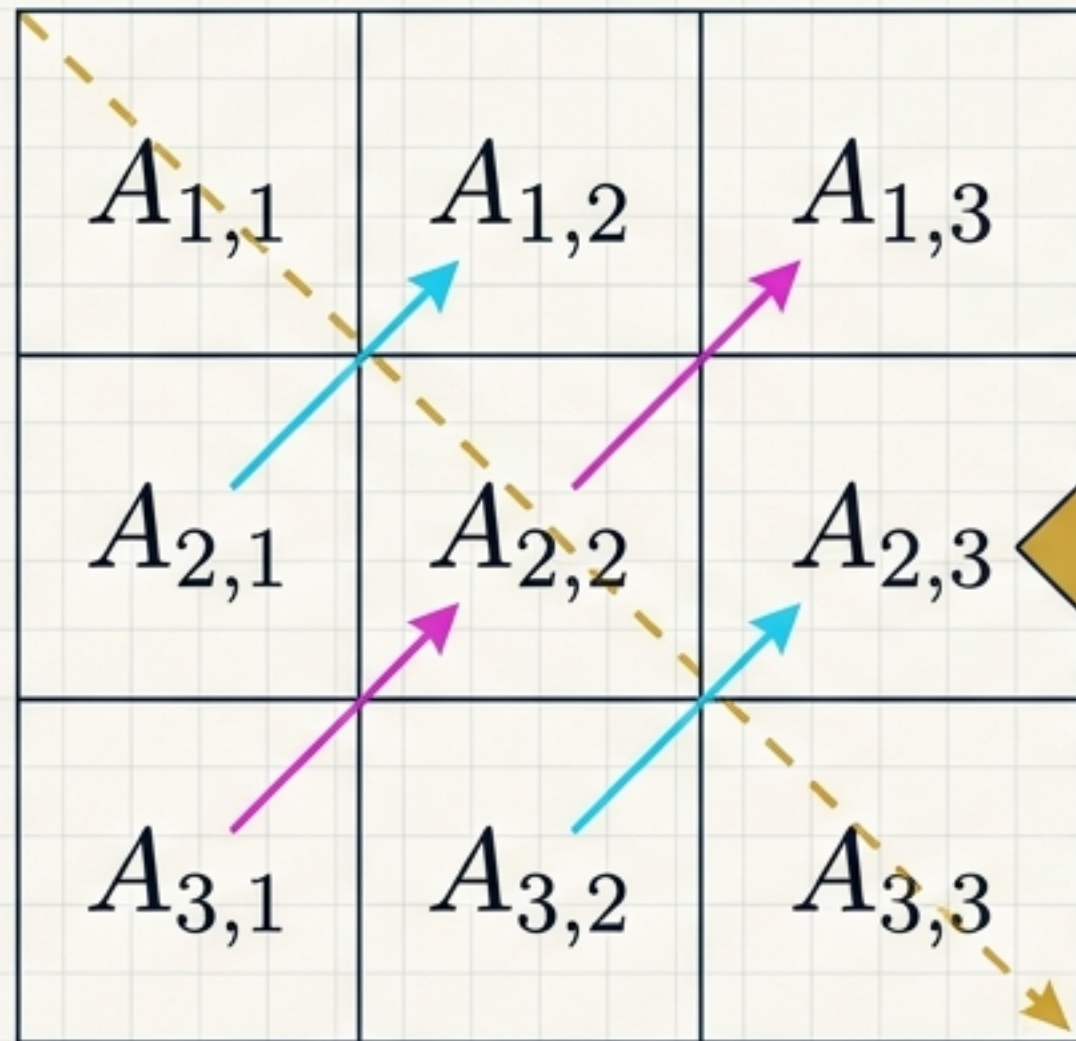
A structural overview from matrix mechanics to formal verification.



From Finite to Infinite Dimensions

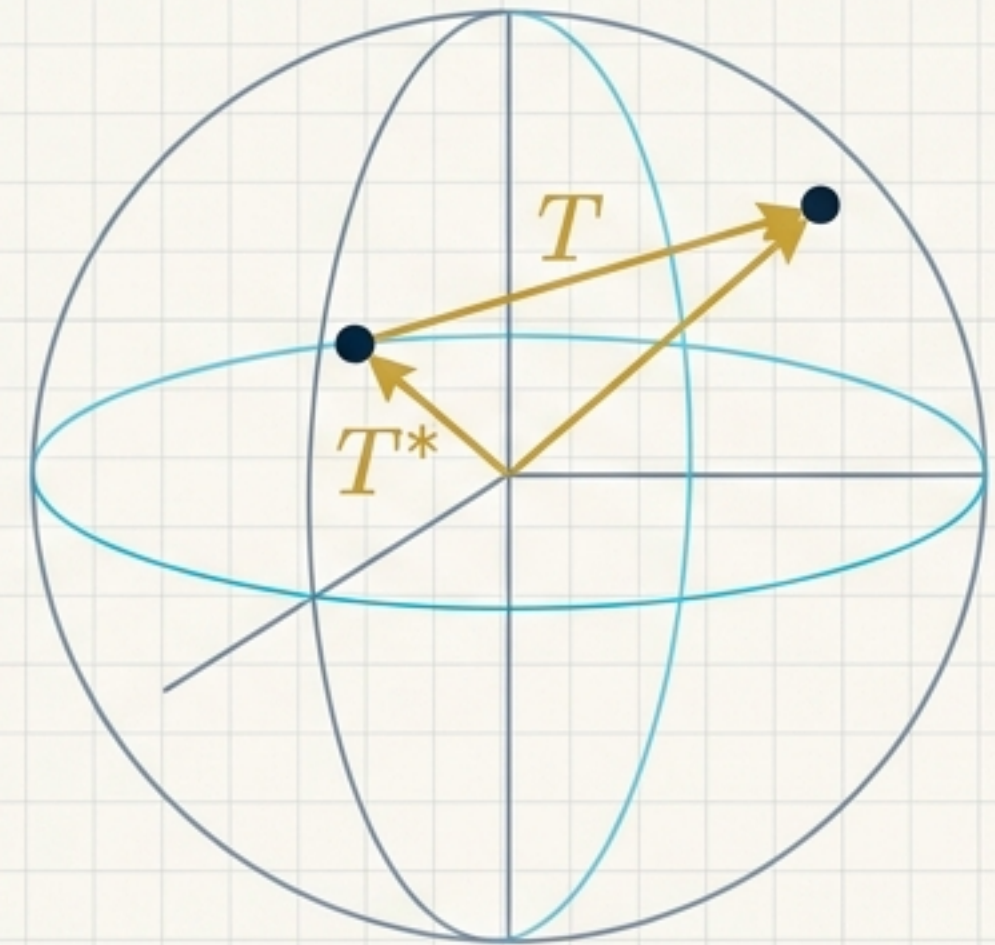
Matrices scale up to Operators on a Hilbert Space. The Adjoint links algebra to geometry.

Finite Dimensions: The Conjugate Transpose



$$a + bi \rightarrow a - bi$$

Infinite Dimensions: The Adjoint



$$\langle Tx, y \rangle = \langle x, T^*y \rangle$$

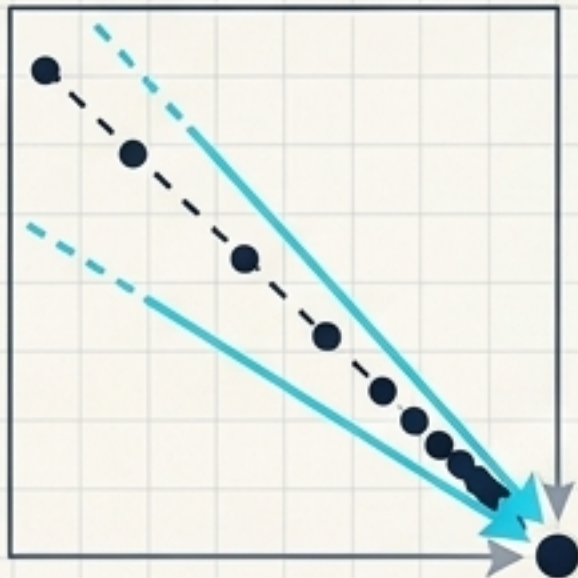
The Arena: What is a C*-Algebra?

A highly rigid environment of operators where geometry and algebra are permanently bound together.

C^* -Algebra

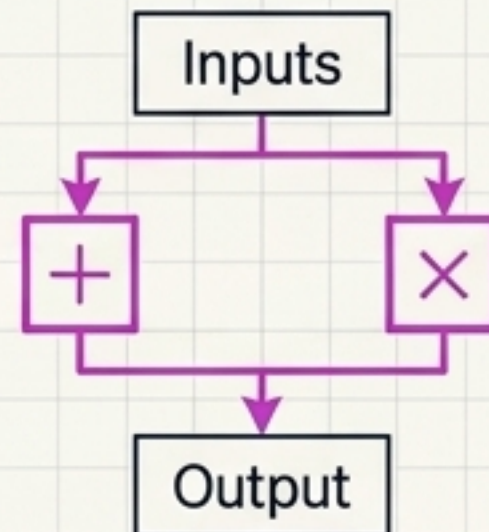
Norm-Closed

Contains its limits.



Algebra

Closed under Sums & Products.



Involution

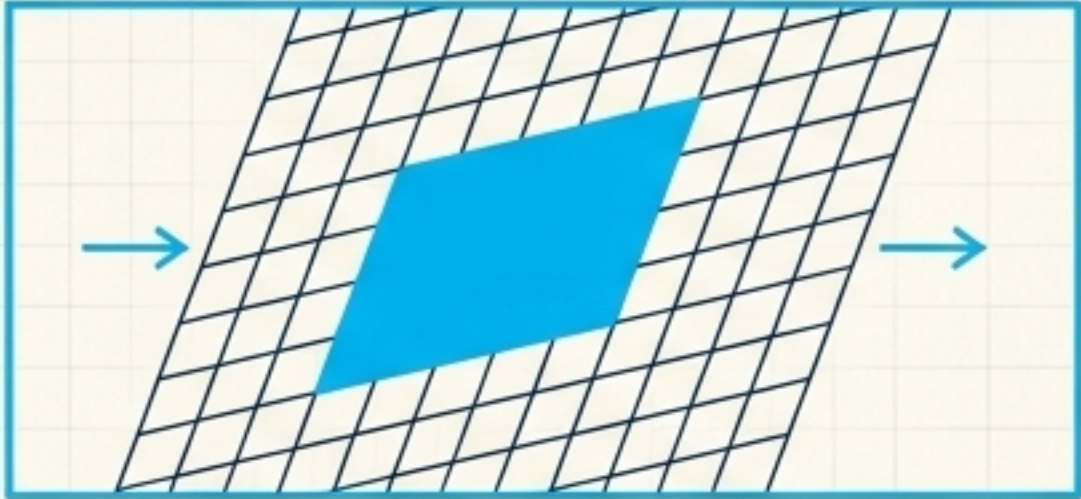
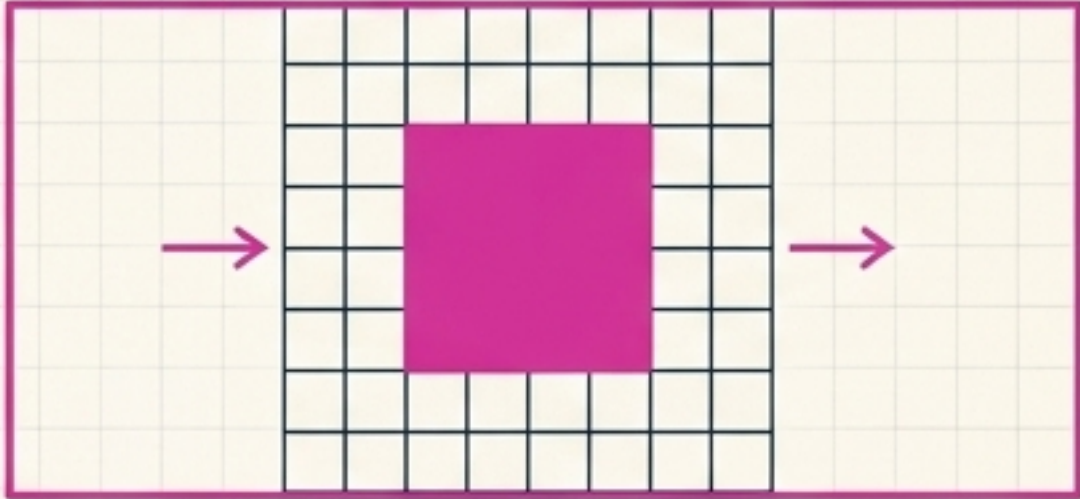
Closed under Adjoints (*).

$$T \xrightarrow{(*)} T^*$$

$$\|x x^*\| = \|x\|^2$$

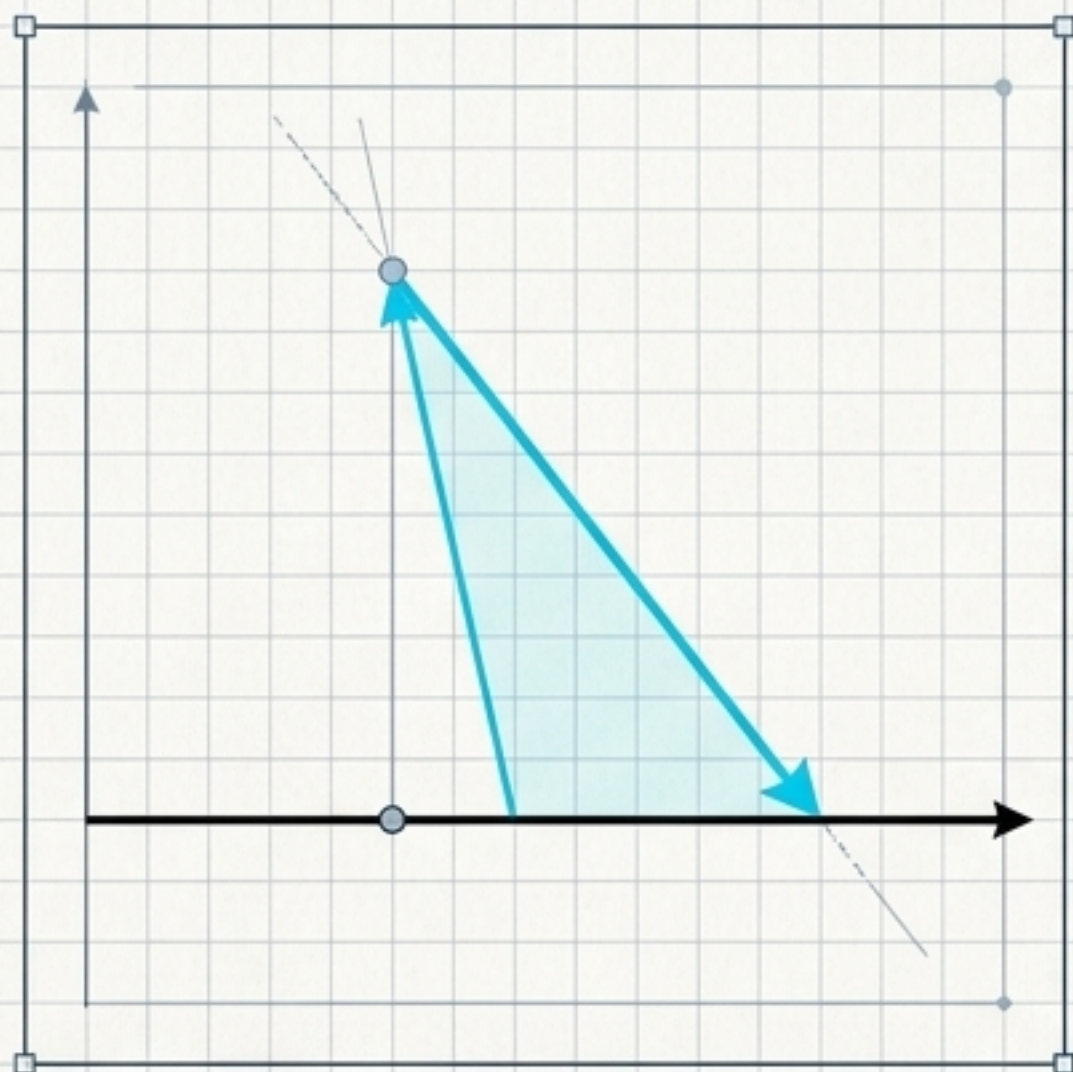
The Core Conflict

Kadison's problem pits a loose algebraic map against a rigid geometric one.

	Bounded Homomorphism (π)	*-Homomorphism (ρ)
		
Respects Sums?	✓ Linearity holds	✓ Linearity holds
Respects Products?	✓ Algebraic structure preserved	✓ Algebraic structure preserved
Respects Adjoints?	✗ Ignores geometry $\pi(x^*) \neq \pi(x)^*$	✓ Respects geometry $\rho(x^*) = \rho(x)^*$

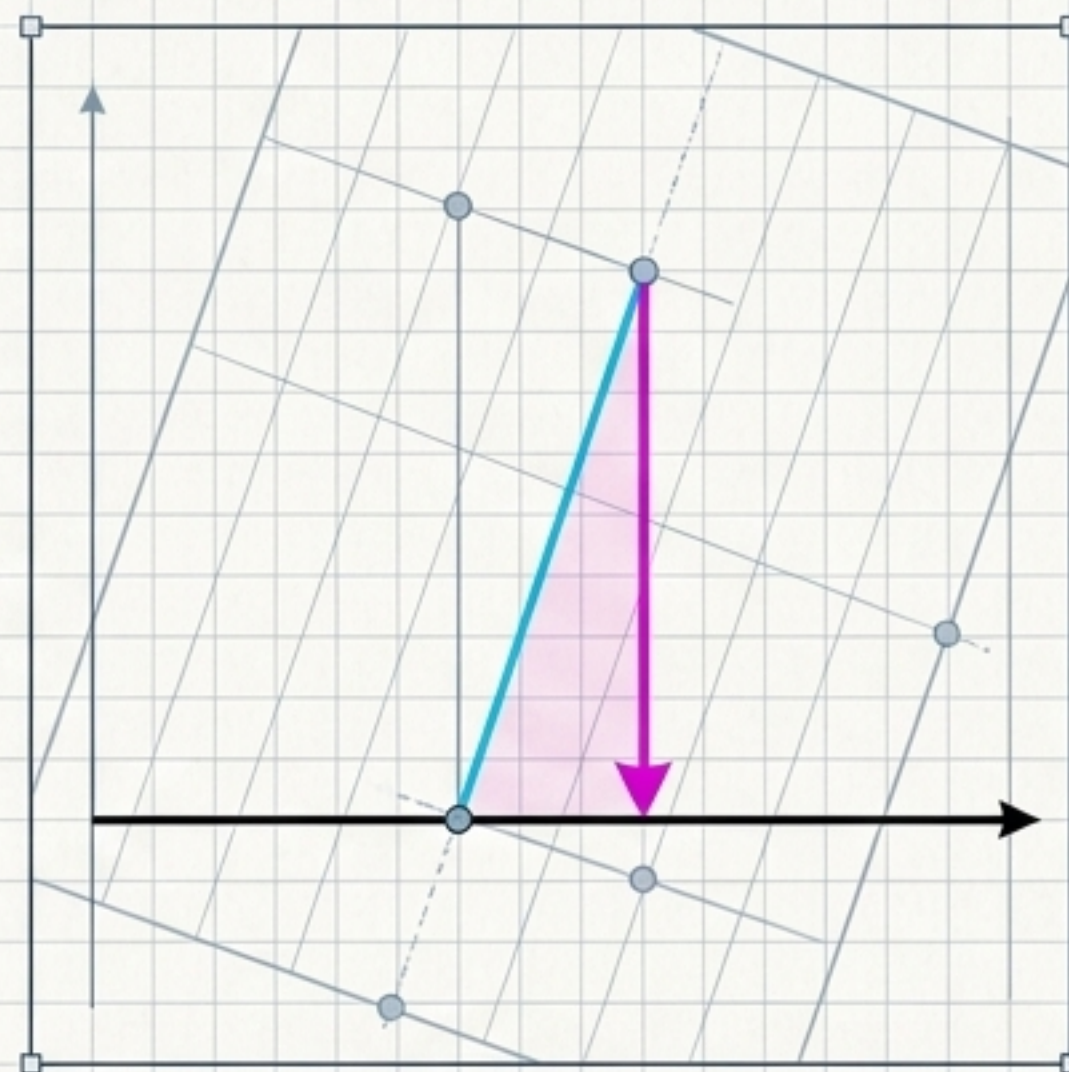
The Fix: Changing the Inner Product

We aren't changing the operator; we are straightening the lens we view it through.



Oblique Projection
Skewed inner product

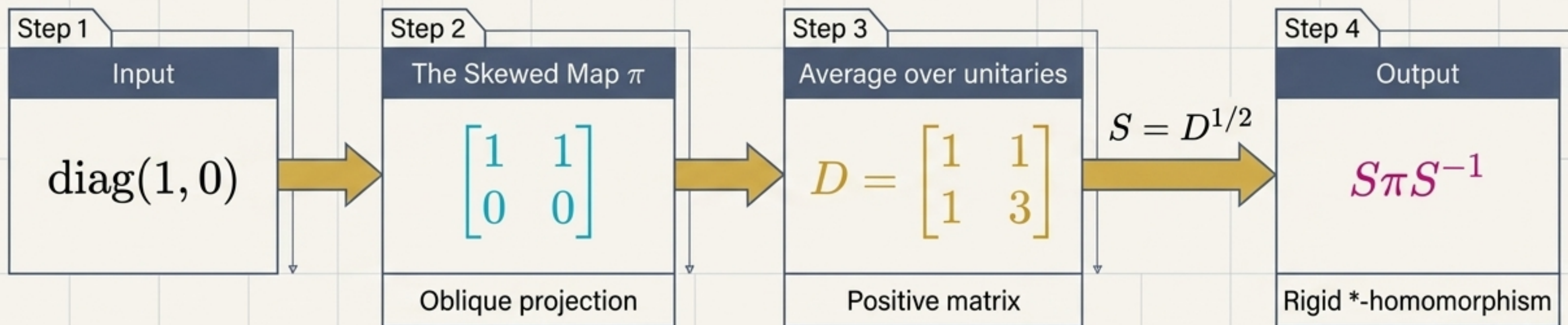
Apply Similarity S :
 $\rho(a) = S \pi(a) S^{-1}$



Orthogonal Projection
Equivalent inner product

Straightening a 2x2 Matrix

A concrete example proving that similarities can fix skewed projections.



$$\text{Condition Bound: } \|S\| \|S^{-1}\| = 1 + \sqrt{2} = \|\pi\|$$

Kadison's 1955 Question

Does this always work?



1955

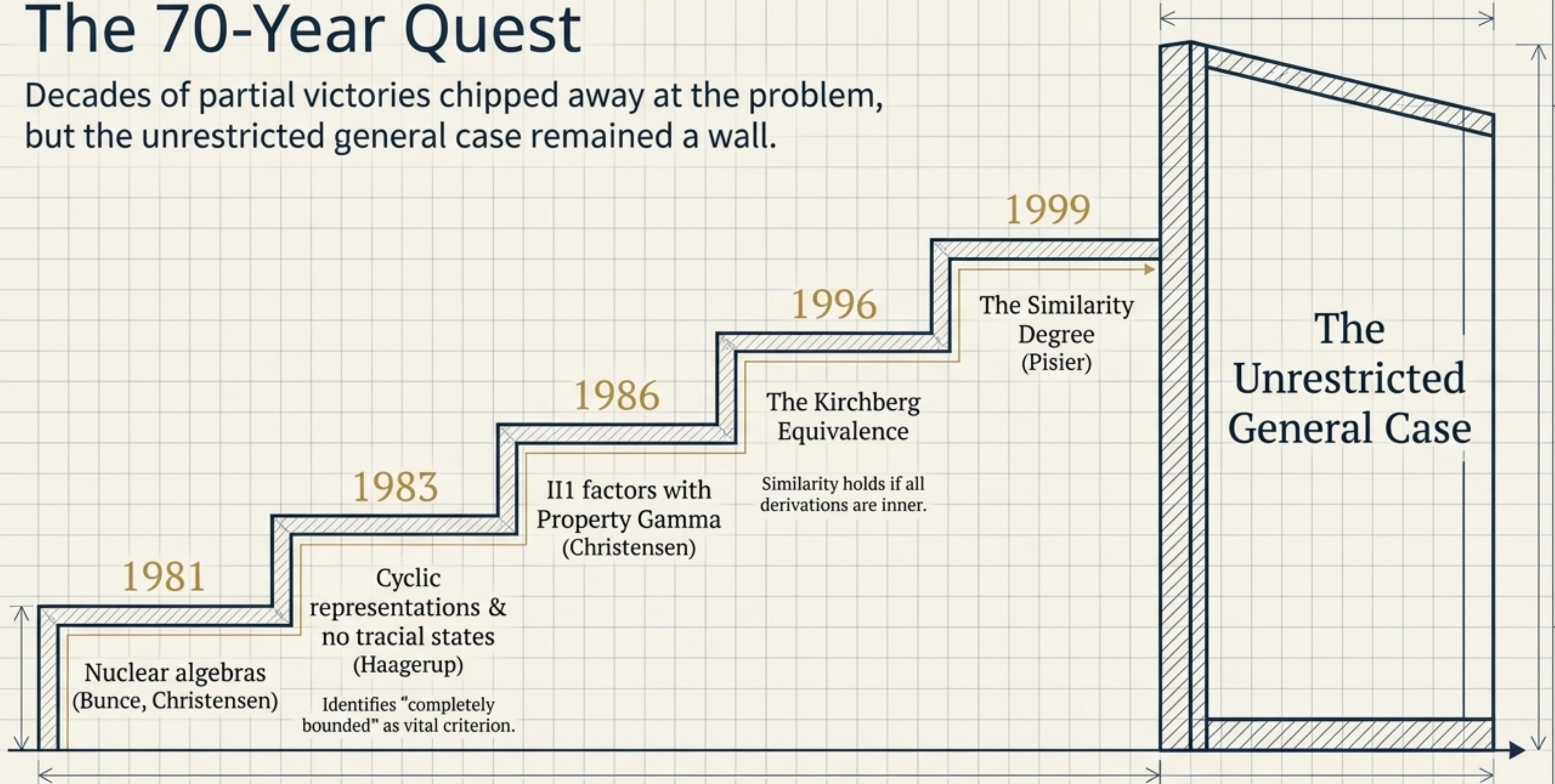
Kadison's question, paraphrased

Can every bounded homomorphism of a C^* -algebra, on any Hilbert space, be straightened into a $*$ -homomorphism by a bounded change of coordinates?

- **Multiplicativity** and **boundedness** alone must produce one equivalent inner product.
- If true in 2×2 matrices, is it true infinitely everywhere?

The 70-Year Quest

Decades of partial victories chipped away at the problem, but the unrestricted general case remained a wall.



The Unrestricted Resolution



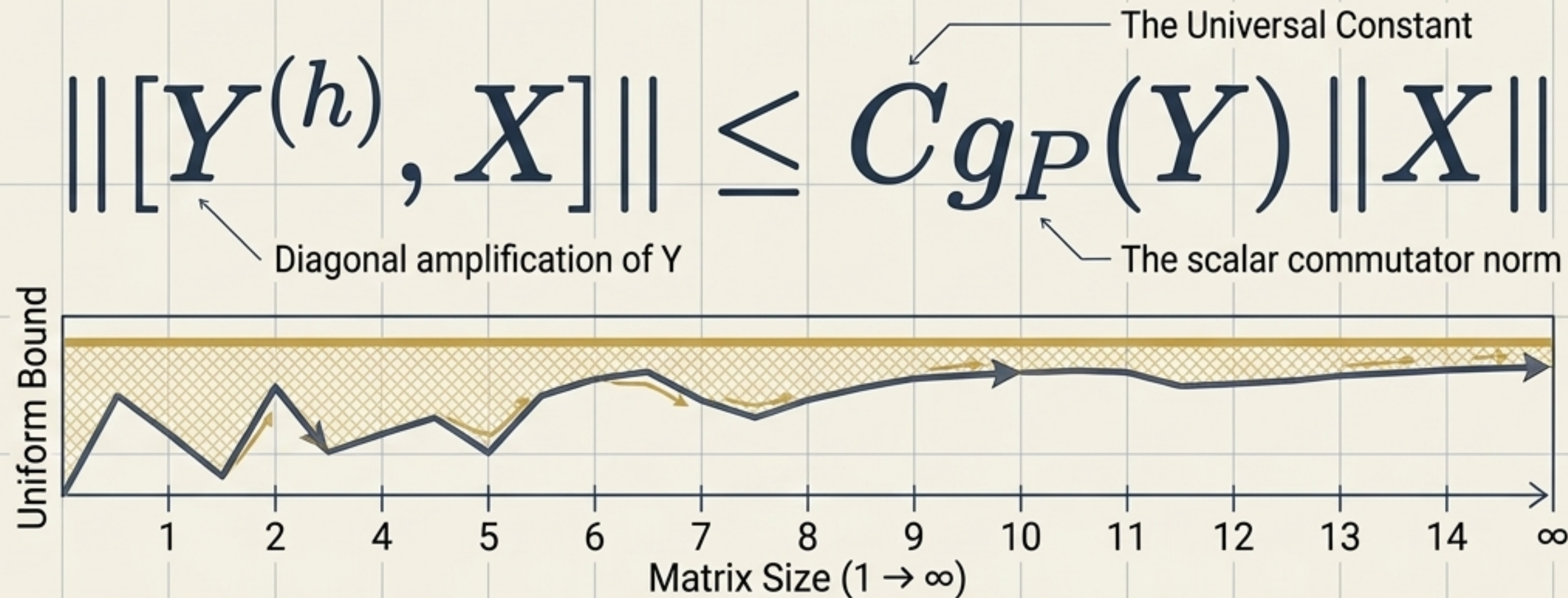
A 2026 paper by OpenAI officially solves Kadison's similarity conjecture positively.

The Theorem: *Every bounded unital algebra homomorphism from a complex C^* -algebra into bounded operators on an arbitrary Hilbert space is similar to a $*$ -homomorphism.*

No Restrictions: No separability, finite generation, or nuclearity conditions required.

The Engine

A uniform commutator estimate independent of matrix size.

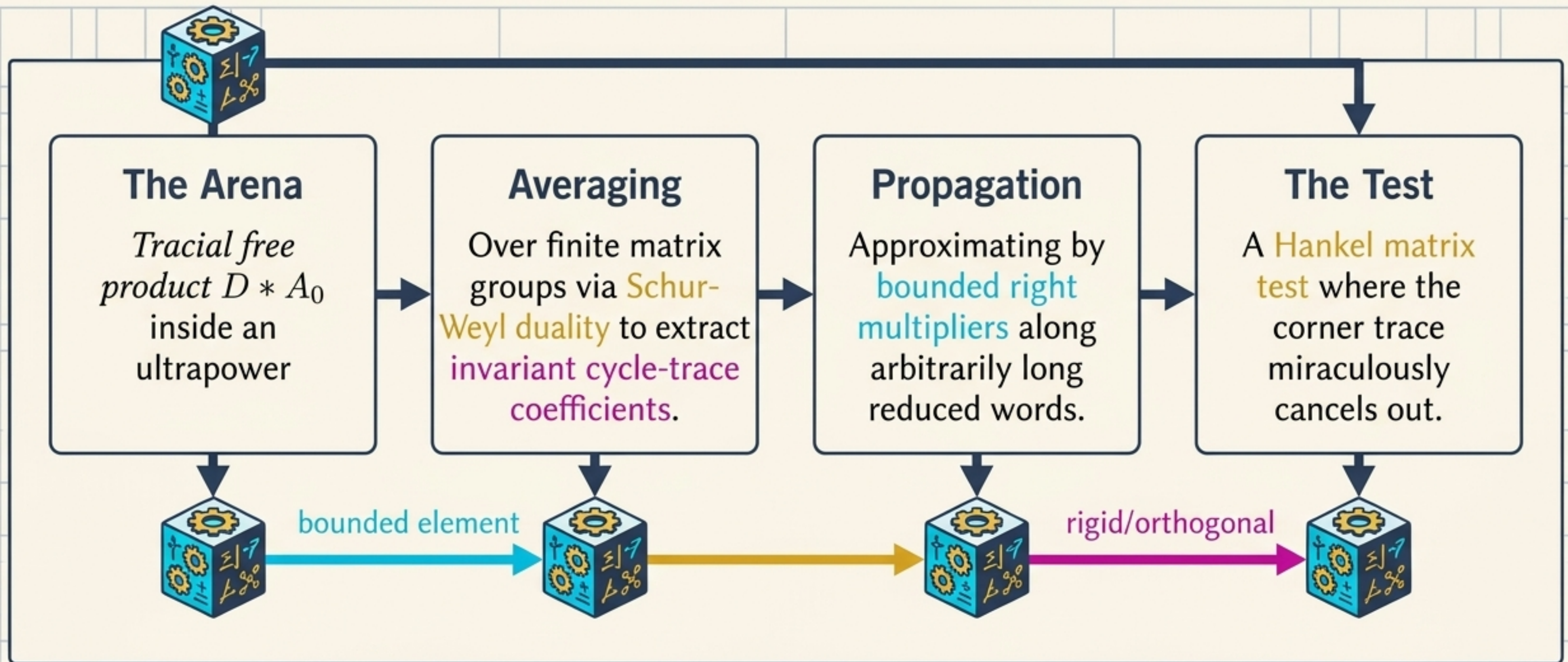


The Constant: $C \approx 1.19$ million

A **massive but finite upper limit** holding the geometry together across all dimensions.

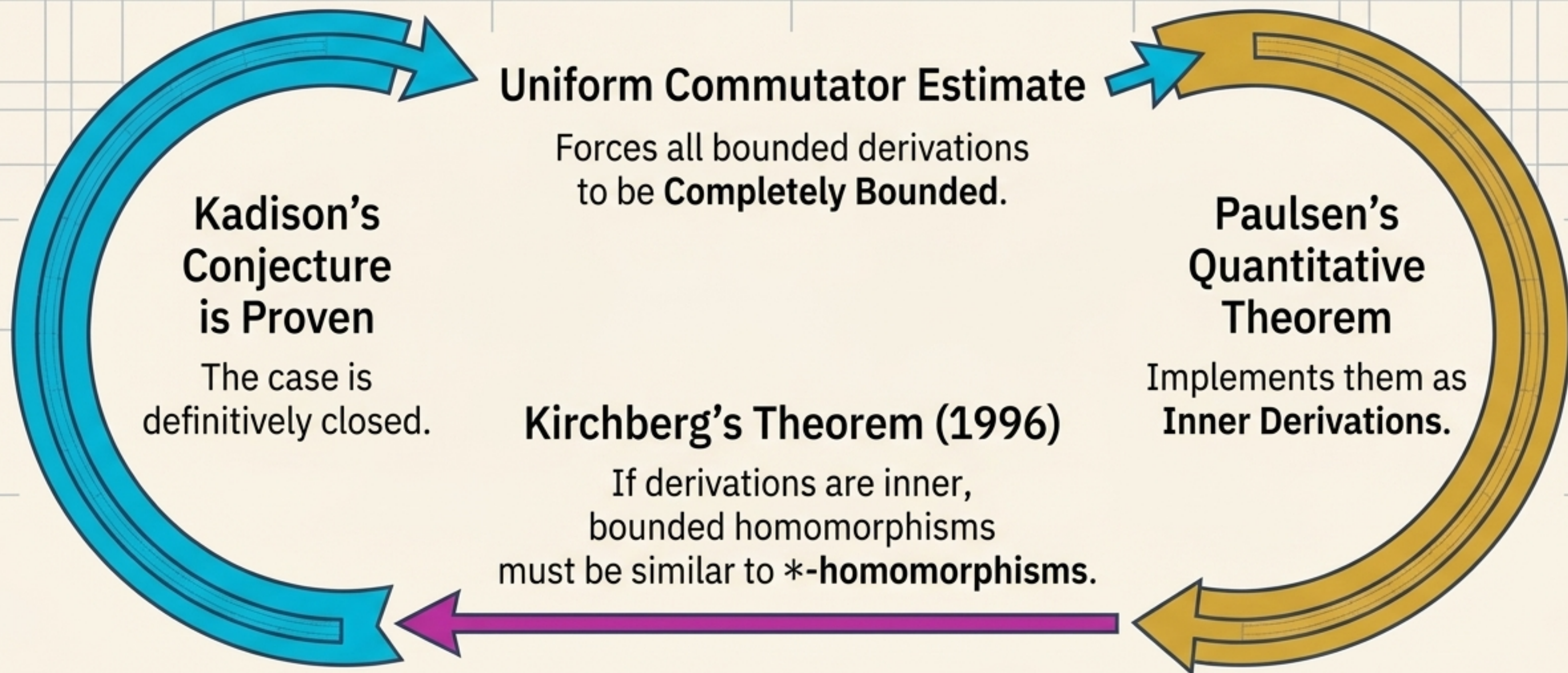
Solving the Hard Case: II_1 Factors

The estimate must be completely uniform, regardless of the factor or representation multiplicity.



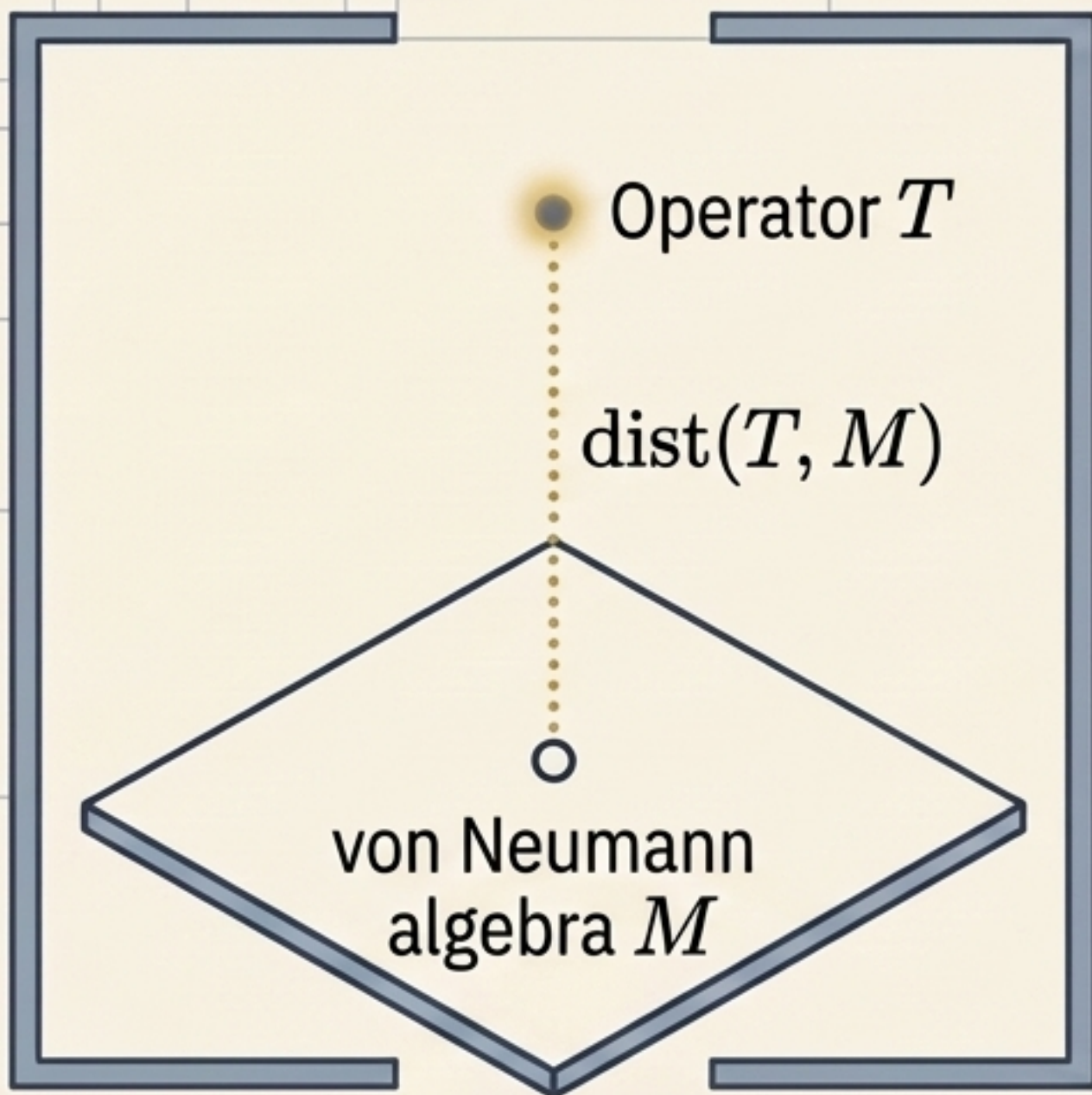
Closing the Loop

Scaling local cyclic-domain estimates back to the historical framework.



The Fallout: Universal Hyperreflexivity

Every unital von Neumann algebra on an arbitrary complex Hilbert space is hyperreflexive.



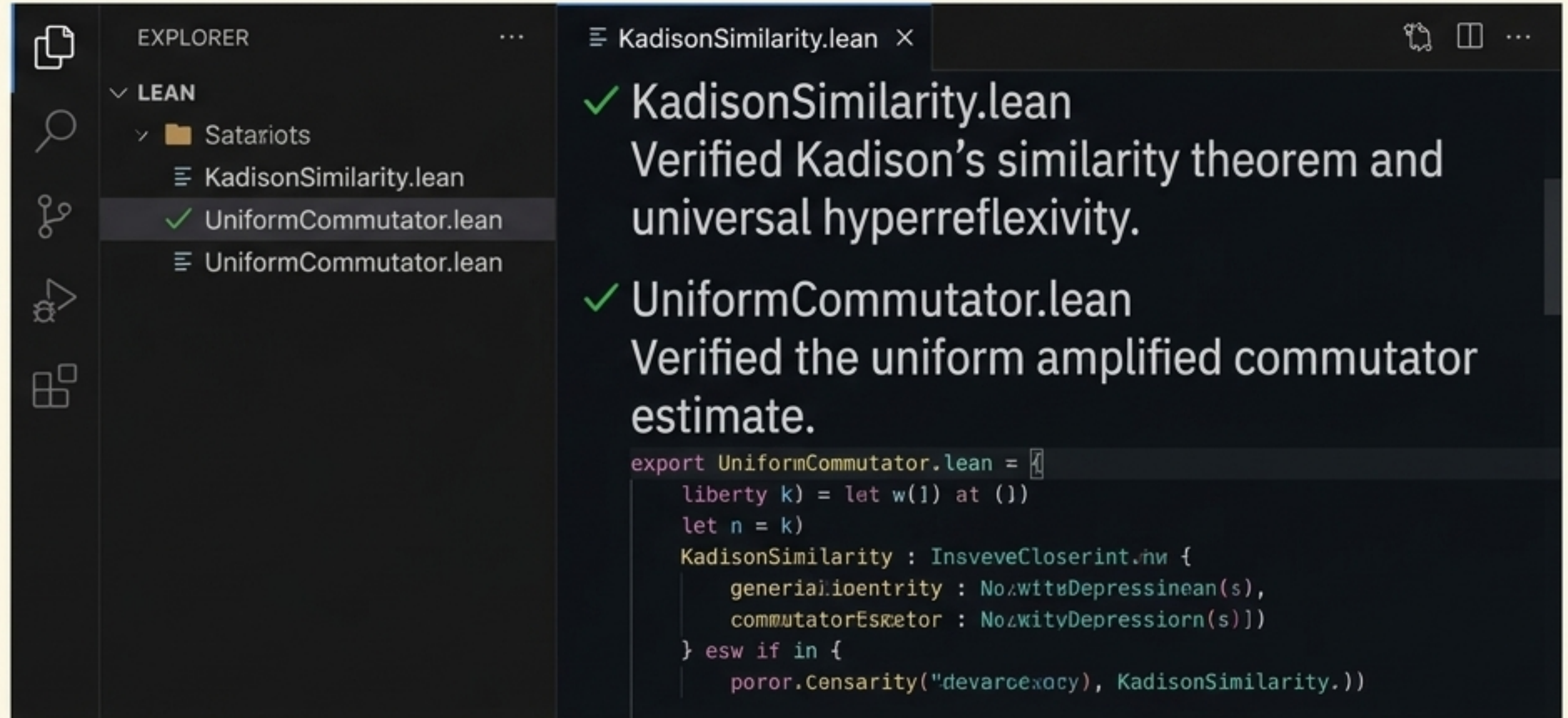
$$\text{dist}(T, M) \leq 2C\alpha_M(T)$$

- The distance from T to M is at most $2C$ times $\alpha_M(T)$, the largest off-diagonal block $(1-e)Te$ over projections e in the commutant M' .
- One universal constant ($2C$) governs the space, completely independent of the algebra or the Hilbert space itself.

Lean Formalization Status

A Lean scope document lists Comparator statements for the main results.

A 70-year
human
question
answered,
checked, and
forever
inscribed in
code.



The screenshot shows the Lean IDE interface. On the left, the 'EXPLORER' panel displays a project structure under 'LEAN' with a folder 'Satarlots' containing two files: 'KadisonSimilarity.lean' and 'UniformCommutator.lean'. Both files are marked with a green checkmark. On the right, the 'Editor' panel shows the 'KadisonSimilarity.lean' file. It contains two entries, each preceded by a green checkmark:

- ✓ KadisonSimilarity.lean
Verified Kadison's similarity theorem and universal hyperreflexivity.
- ✓ UniformCommutator.lean
Verified the uniform amplified commutator estimate.

Below these entries, a snippet of Lean code is visible:

```
export UniformCommutator.lean =  
  liberty k) = let w(1) at ()  
  let n = k)  
  KadisonSimilarity : InsveveClosierint.nw {  
    generialioentryty : NozwitwDepressinean(s),  
    commutatorEskeator : NozwityDepressiorn(s))  
  } esw if in {  
    poror.Censarity("devarceexocy), KadisonSimilarity.))
```

The Lean build was not re-run for this explainer; the paper is an unrefereed preprint.